

**Course: Fundamentals of Generative AI and Large Language
Models: Theory and Practice**

**Week 3: Variational Autoencoders (VAEs):
Foundations**

Today's Session Overview

- Self-information, entropy, cross-entropy, and the detailed derivation of KL divergence

NPTTEL

Self-information

Self-information: measures how surprising one particular outcome is.

For an outcome x with probability $P(x)$, self-information is defined as:

$$I(x) = \log_2 \frac{1}{P(x)}$$

This is also written as:

$$I(x) = -\log_2 P(x)$$

because

$$\log_2 \frac{1}{P(x)} = -\log_2 P(x)$$

Suppose two independent events happen:

log converts multiplication of probabilities into addition of information.

$$\log(ab) = \log a + \log b \quad \checkmark$$

- ✓ If $P(x)$ is large, then $I(x)$ is small (Less surprise)
- ✓ If $P(x)$ is small, then $I(x)$ is large (More surprise)

That is exactly why the formula uses $\frac{1}{P(x)}$

When $\log_2$ is used, self-information is measured in **bits**.

$$P(\text{Sunny}) = 0.875 \quad \checkmark$$

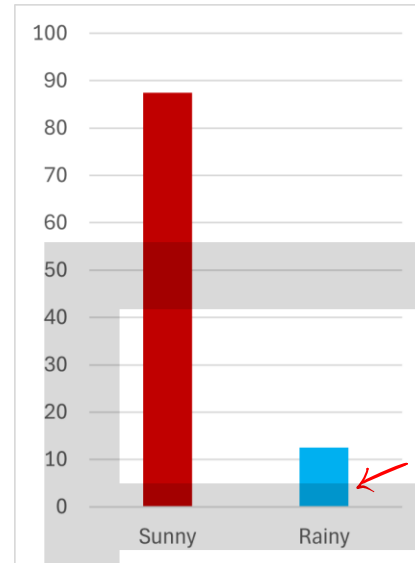
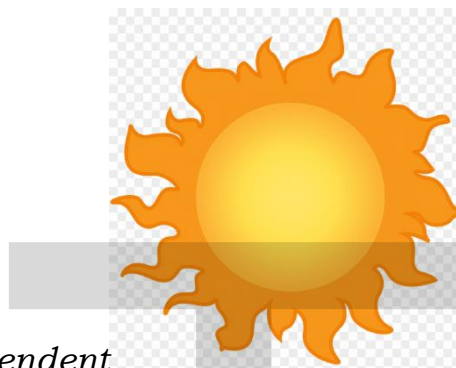
$$I(\text{Sunny}) = -\log_2(0.875) \approx \underline{0.193} \text{ bits} \quad \checkmark$$

Rain carries more self-information because it was less likely.

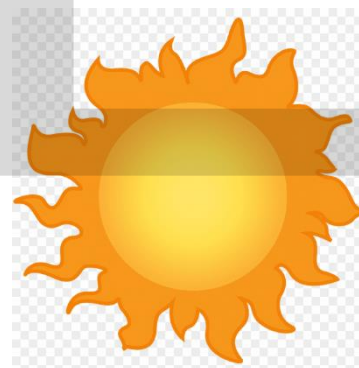
$$P(\text{Rainy}) = 0.125 \quad \checkmark$$

$$I(\text{Rainy}) = -\log_2(0.125) = \underline{3} \text{ bits} \quad \uparrow$$

We want to check tomorrow's weather



If tomorrow is sunny, you are not very surprised because sunny weather was highly likely.



If tomorrow is rainy, you are more surprised because rain was unlikely.

So self-information measures the surprise of **one outcome only**.

Example credit to Jia-Bin Huang

Entropy

Self-information:

$$I(x) = -\log_2 P(x)$$

This gives the surprise associated with **one particular outcome**.

Let

$X = \text{tomorrow's weather}$

The possible outcomes are:

$X \in \{\text{Sunny}, \text{Rainy}\}$

Before tomorrow arrives, we do not know which outcome will occur. There is some uncertainty. X could become either:

$X = \text{Sunny}$

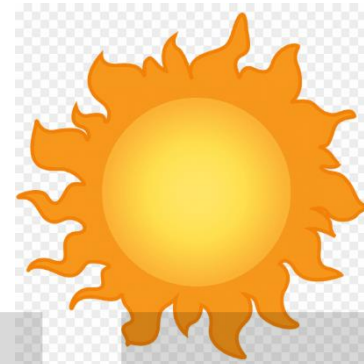
or

$X = \text{Rainy}$

The model's predicted probabilities :

$P(X = \text{Sunny}) = 0.875$

$P(X = \text{Rainy}) = 0.125$



87.50% ✓



12.50% ✓

The model is saying:

87.5% chance of sunny weather

12.5% chance of rainy weather

Together, these form the model's predicted probability distribution:

$P(X) = [0.875, 0.125]$

Entropy considers all possible outcomes of X and their probabilities.

Entropy **measures this overall uncertainty**.

Entropy

Entropy is used because **self-information describes only one observed outcome**, while we often want one number that describes the **uncertainty of the entire probability distribution**.

For one outcome x ,

$$I(x) = -\log_2 P(x)$$

But before the outcome is known, several outcomes are possible. Entropy combines the surprise of all outcomes into one average value.

$$H(P) = \sum_x P(x) I(x)$$

Therefore:

$$H(P) = \sum_x P(x) [-\log_2 P(x)] = - \sum_x P(x) \log_2 P(x)$$

$$H(P) = \sum_x P(x) \log_2 \frac{1}{P(x)}$$

Entropy multiplies the surprise of each outcome by its probability and adds the results.

∴, Before observing the outcome, how uncertain is the whole situation on average? - Entropy



87.50%



12.50%

$$H(P) = 0.875 \log_2 \frac{1}{0.875} + 0.125 \log_2 \frac{1}{0.125} = 0.544$$

Because one outcome is much more likely than the other, the model is fairly confident, so the entropy is relatively low.

0.50

0.50

$H(P) = 1$, Maximum uncertainty. The result is most unpredictable.

0

100

$H(P) = 0$, No uncertainty. Then the weather is completely predictable.

Cross-entropy Example

Suppose the true outcome is Sunny.

The target distribution is:

$$P = [1, 0] = [\text{Sunny}, \text{Rainy}]$$

Assume the model predicts:

$$Q = [0.875, 0.125]$$

Then the cross-entropy is:

$$H(P, Q) = -[1 \log_2(0.875) + 0 \log_2(0.125)]$$

$$H(P, Q) = -\log_2(0.875)$$

$$H(P, Q) \approx 0.193$$

The loss is small because the model assigned a high probability, 0.875, to the correct outcome Sunny.

$P(1, 0)$

Suppose the actual outcome is still Sunny, but the model predicts:

$$Q = [\overset{S}{0.125}, \overset{R}{0.875}]$$

Then:

$$H(P, Q) = -[1 \log_2(0.125) + 0 \cancel{\log_2(0.875)}]$$

$$H(P, Q) = -\log_2(0.125)$$

$$H(P, Q) \approx \underline{\underline{3}}$$

The loss is large because the model assigned only 0.125 probability to the correct class.

So:

Cross-entropy is small when the model assigns high probability to the correct outcome and large when it assigns low probability to the correct outcome.

Cross-entropy

Cross-entropy measures how well the predicted distribution Q matches the target distribution P .

For one outcome x , the model assigns probability $Q(x)$.

The surprise according to the model is:

$$I_Q(x) = -\log_2 Q(x)$$

Cross-entropy checks how surprised the model is about the correct answer.

$$H(P, Q) = \sum_x P(x) I_Q(x)$$

Therefore:

$$H(P, Q) = -\sum_x P(x) \log_2 Q(x)$$

$P(x)$ = true probability of outcome x

$I_Q(x) = -\log_2 Q(x)$ = surprise assigned to that outcome by the model

Therefore:

$$P(x) I_Q(x)$$

$$D_{KL}(P \parallel Q) = H(P, Q) - H(P)$$

$$D_{KL}(P \parallel Q) = \left(- \sum_x P(x) \log_2 Q(x) \right) - \left(- \sum_x P(x) \log_2 P(x) \right)$$

$$D_{KL}(P \parallel Q) = - \sum_x P(x) \log_2 Q(x) + \sum_x P(x) \log_2 P(x)$$

Rearrange:

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log_2 P(x) - \sum_x P(x) \log_2 Q(x)$$

Take common term $P(x)$

$$D_{KL}(P \parallel Q) = \sum_x P(x) (\log_2 P(x) - \log_2 Q(x))$$

Cross-entropy:

$$H(P, Q) = - \sum_x P(x) \log_2 Q(x)$$

Entropy:

$$H(P) = - \sum_x P(x) \log_2 P(x)$$

$$\log a - \log b = \log \frac{a}{b}$$

Substitute:

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)}$$

Next Session

- Introduction to Variational Autoencoders (VAEs), followed by a detailed discussion of the VAE encoder.

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Thank you